Week 7 at a glance

We will be learning and practicing to:

- Model systems with tools from discrete mathematics and reason about implications of modelling choices. Explore applications in CS through multiple perspectives, including software, hardware, and theory.

M – Defining important sets of numbers, e.g. set of integers, set of rational numbers

F – Defining functions using multiple representations

F – Determining whether a given function is one-to-one, onto, and/or invertible

F – Using functions to compare cardinality of sets

- Clearly and unambiguously communicate computational ideas using appropriate formalism. Translate across levels of abstraction.
  - Using appropriate signpost words to improve readability of proofs, including 'arbitrary' and 'assume'

- Know, select and apply appropriate computing knowledge and problem-solving techniques. Reason about computation and systems. Use mathematical techniques to solve problems. Determine appropriate conceptual tools to apply to new situations. Know when tools do not apply and try different approaches. Critically analyze and evaluate candidate solutions.
  - Determining what evidence is required to establish that a quantified statement is true or false

F – Evaluating quantified statements about finite and infinite domains

- Apply proof strategies, including direct proofs and proofs by contradiction, and determine whether a proposed argument is valid or not.
  - Identifying the proof strategies used in a given proof
  - Identifying which proof strategies are applicable to prove a given compound proposition based on its logical structure
  - Carrying out a given proof strategy to prove a given statement

M – Tracing and/or modifying a proof by contradiction

  - Using proofs as knowledge discovery tools to decide whether a statement is true or false

TODO: Discussion this evening!

See Piazza for office hours poll

Review quiz based on Week 6 class material (due Monday)

No class Wednesday of Week 7 in observance of UCSD holiday.

Midquarter feedback: please let us know what's working well for you and what isn't. `https://canvas.ucsd.edu/courses/71479/quizzes`

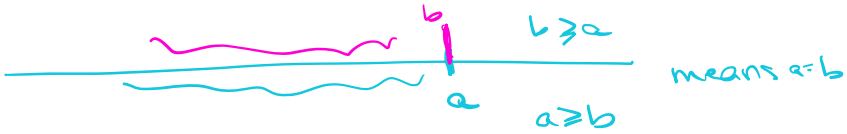
50 responses so far!

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Week 7 Monday: Proof by Contradiction

"Want to show that it's impossible for statement we care about to be false".

New! Proof by Contradiction
To prove that a statement p is true, pick another statement r and once we show that $\neg p \to (r \wedge \neg r)$ then we can conclude that p is true.
Informally The statement we care about can't possibly be false, so it must be true.



Least and greatest

For a set of numbers X, how do you formalize "there is a greatest X" or "there is a least X"?

There is a greatest X: $\exists a \in X \; \forall b \in X \; (b \leq a)$
There is a least X: $\exists a \in X \; \forall b \in X \; (a \leq b)$

Prove ~~or disprove~~: There is a least prime number.

$X = \{n \in \mathbb{Z} \mid n \text{ is prime}\}$

Need witness, value for $a = 2$
WTS $\forall b \in$ Primes $(2 \leq b)$.
Because 2 is prime, and universal holds, because for arbitrary e a prime, it's defined to be int greater than 1 so is greater than or equal to 2 ☒

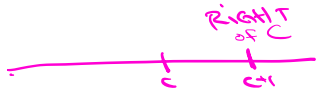
~~Prove~~ or **disprove**: There is a greatest integer. $X = \mathbb{Z}$

Approach 1, De Morgan's and universal generalization:

WTS $\forall a \in \mathbb{Z} \; \exists b \in \mathbb{Z} \; (a < b)$
Towards universal generalization, consider arbitrary integer, a, WTS $\exists b \in \mathbb{Z} \; (a < b)$. Consider $b = a+1$. Then b is an integer and $b > a$, so it's a witness ☺

Approach 2, proof by contradiction:

WTS "There isn't a greatest integer"
Towards a contradiction, assume there is a greatest integer. From assumption, let c be witness greatest integer.
Know that c is integer.
So $c+1$ is also integer.
By properties of numbers $c+1 > c$. (r)



But since c is supposed to be greatest integer (and $c+1$ is an integer) $c \geq c+1$, in other words $\neg(c+1 > c)$. ($\neg r$)



So we showed that the negation of original statement meant that $r \wedge \neg r$, a contradiction. So we conclude that the original statement must be true ☒

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Extra examples: Prove or disprove that $\mathbb{N}$, $\mathbb{Q}$ each have a least and a greatest element.

Definition: **Greatest common divisor** Let a and b be integers, not both zero. The largest integer d such that d is a factor of a and d is a factor of b is called the greatest common divisor of a and b and is denoted by $gcd(\ (a,b)\ )$.

Why do we restrict to the situation where a and b are not both zero?

Calculate $gcd(\ (10,15)\ )$ $= 5$

Positive factors of 10 : 1, 2, 5, 10

Positive factors of 15 : 1, 3, 5, 15

Calculate $gcd(\ (10,20)\ )$ $= 2^{\min(1,2)} \cdot 5^{\min(1,1)} = 2^1 \cdot 5^1 = 10$

$10 = 2^1 \cdot 5^1$

$20 = 2^2 \cdot 5^1$

Claim: For any integers a, b (not both zero), $gcd(\ (a,b)\ ) \geq 1$.

Proof: *Show that* 1 *is a common factor of any two integers, so since the gcd is the greatest common factor it is greater than or equal to any common factor.*

Let a, b be arbitrary integers. Assume (towards a direct proof) that $\neg(a=0 \wedge b=0)$. WTS $\gcd((a,b)) \geq 1$. By assumption, $\gcd((a,b))$ is defined. Also, 1 is a factor of a (as witnessed by the integer a, since $a = 1 \cdot a$) and 1 is also a factor of b (as witnessed by the integer b, since $b = 1 \cdot b$). Thus 1 is a common divisor of a and b. Since $\gcd((a,b))$ is the greatest common divisor, we must have that $1 \leq \gcd((a,b))$. ☺

Claim: For any positive integers a, b, $gcd(\ (a,b)\ ) \leq a$ and $gcd(\ (a,b)\ ) \leq b$.

Proof *Using the definition of gcd and the fact that factors of a positive integer are less than or equal to that integer.*

WTS $\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+ (\gcd((a,b)) \leq a \wedge \gcd((a,b)) \leq b)$

Let a, b be arbitrary positive integers. We have two subgoals.

Subgoal ① WTS $\gcd((a,b)) \leq a$. By definition of gcd, $\gcd((a,b))$ is a factor of a and is positive, so by one of the statements proved last week, $\gcd((a,b)) \leq a$.

Subgoal ② WTS $\gcd((a,b)) \leq b$. The identical argument to what we just used for subgoal ① works, replacing a by b. ☺

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Claim: For any positive integers a, b, if a divides b then $gcd(\ (a,b)\ ) = a$.

Proof *Using previous claim and definition of gcd.*

Towards universal generalization, let a and b be positive integers. Towards direct proof, assume a divides b. In other words, a is a divisor of b. Notice that b is also a divisor of b (we proved last week that all nonzero integers divide themselves) so a is a common divisor of a and b. By definition of gcd therefore, $a \leq gcd((a,b))$. Also, since gcd((a,b)) is a factor of a and both a and gcd((a,b)) are positive, $gcd((a,b)) \leq a$. By properties of numbers, now that we know $a \leq gcd((a,b))$ and $gcd((a,b)) \leq a$, we can conclude $a = gcd((a,b))$. ∎

Claim: For any positive integers a, b, c, if there is some integer q such that $a = bq + c$,

$$gcd(\ (a,b)\ ) = gcd(\ (b,c)\ )$$

Proof *Prove that any common divisor of a, b divides c and that any common divisor of b, c divides a.*

Let a, b, c be arbitrary positive integers. Assume, towards direct proof, that there is integer q with $a = bq + c$. WTS $gcd((a,b)) = gcd((b,c))$. Equivalently, WTS $gcd((a,b)) \leq gcd((b,c))$ and $gcd((b,c)) \leq gcd((a,b))$. For each subgoal, it's enough to show that the LHS is a common divisor of the numbers whose gcd is on the RHS.

Subgoal ① WTS gcd((a,b)) is a factor of both b and c. By definition, it's a factor of b. To prove it's also a factor of c, we rewrite the assumption $a = bq + c$ as $c = a - bq$. Define x and y to be the integers we multiply gcd((a,b)) by to get a and b (respectively). Then $c = x\,gcd((a,b)) - y\,gcd((a,b))\,q = (x - yq)\,gcd((a,b))$. Since $x - yq$ is an integer, we have that gcd((a,b)) is a factor of c. ∎

Subgoal ② WTS gcd((b,c)) is a common divisor of a and b. A similar argument works [details left as extra practice.] ∎

Lemma: For any integers p, q (not both zero), $gcd\left(\left(\frac{p}{gcd(\ (p,q)\ )}, \frac{q}{gcd(\ (p,q)\ )}\right)\right) = 1$. In other words, can reduce to relatively prime integers by dividing by gcd.

Proof: means gcd( ) = 1

Let x be arbitrary positive integer and assume that x is a factor of each of $\frac{p}{gcd(\ (p,q)\ )}$ and $\frac{q}{gcd(\ (p,q)\ )}$. This gives integers α, β such that

$$\alpha x = \frac{p}{gcd(\ (p,q)\ )} \qquad \beta x = \frac{q}{gcd(\ (p,q)\ )}$$

Multiplying both sides by the denominator in the RHS:

$$\alpha x \cdot gcd(\ (p,q)\ ) = p \qquad \beta x \cdot gcd(\ (p,q)\ ) = q$$

In other words, $x \cdot gcd(\ (p,q)\ )$ is a common divisor of p, q. By definition of gcd, this means

$$x \cdot gcd(\ (p,q)\ ) \leq gcd(\ (p,q)\ )$$

and since $gcd(\ (p,q)\ )$ is positive, this means, $x \leq 1$.

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Sets of numbers

We've seen multiple representations of the set of positive integers (using base expansions and using prime factorization). Now we're going to expand our attention to other sets of numbers as well.

The **set of rational numbers**, $\mathbb{Q}$ is defined as

$$\left\{\frac{p}{q} \mid p \in \mathbb{Z} \text{ and } q \in \mathbb{Z} \text{ and } q \neq 0\right\} \quad \text{or, equivalently,} \quad \{x \in \mathbb{R} \mid \exists p \in \mathbb{Z} \exists q \in \mathbb{Z}^+ (p = x \cdot q)\}$$

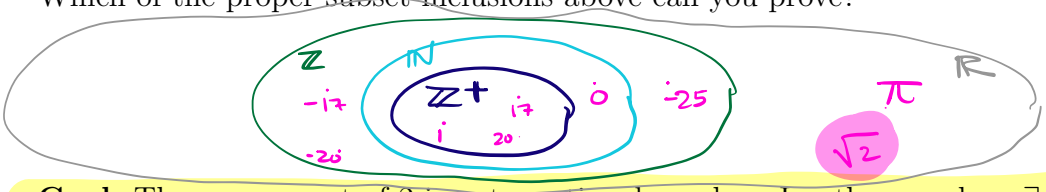
does not require lowest terms.

Extra practice: Use the definition of set equality to prove that the definitions above give the same set.

We have the following subset relationships between sets of numbers:

$$\mathbb{Z}^+ \subsetneq \mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R}$$

Which of the proper subset inclusions above can you prove?



Goal: The square root of 2 is not a rational number. In other words: $\neg\exists x \in \mathbb{Q}(x^2 - 2 = 0)$

Attempted proof: The definition of the set of rational numbers is the collection of fractions p/q where p is an integer and q is a nonzero integer. Looking for a **witness** p and q, we can write the square root of 2 as the fraction $\sqrt{2}/1$, where 1 is a nonzero integer. Since the numerator is not in the domain, this witness is not allowed, and we have shown that the square root of 2 is not a fraction of integers (with nonzero denominator). Thus, the square root of 2 is not rational.

The problem in the above attempted proof is that it proves the chosen witness doesn't work but that's not enough to rule out other witnesses.

Lemma 1: For every two integers a and b, not both zero, with $gcd(\ (a, b)\ ) = 1$, it is not the case that both a is even and b is even.

Proofs in Review Quizzes!

Lemma 2: For every integer x, x is even if and only if x^2 is even.

Proof: Towards a proof by contradiction, we will define a statement r such that $\sqrt{2} \in \mathbb{Q} \to (r \wedge \neg r)$.

Assume that $\sqrt{2} \in \mathbb{Q}$. Namely, there are positive integers p, q such that

$$\sqrt{2} = \frac{p}{q}$$

Let $a = \frac{p}{gcd(\ (p,q)\ )}$, $b = \frac{q}{gcd(\ (p,q)\ )}$, then

$$\sqrt{2} = \frac{a}{b} \quad \text{and} \quad gcd(\ (a, b)\ ) = 1$$

fraction in lowest term

By Lemma 1, a and b are not both even. We define r to be the statement "a is even and b is even", and we have proved $\neg r$.

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Squaring both sides and clearing denominator: $2b^2 = a^2$.

By definition of even, since b^2 is an integer, a^2 is even.

By Lemma 2, this guarantees that a is even too. So, by definition of even, there is some integer (call it c), such that $a = 2c$.

Plugging into the equation:

$$2b^2 = a^2 = (2c)^2 = 4c^2$$

and dividing both sides by 2

$$b^2 = 2c^2$$

and since c^2 is an integer, b^2 is even. By Lemma 2, b is even too. Thus, a is even and b is even and we have proved r.

In other words, assuming that $\sqrt{2} \in \mathbb{Q}$ guarantees $r \wedge \neg r$, which is impossible, so $\sqrt{2} \notin \mathbb{Q}$. QED

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Week 7 Friday: Cardinality and functions

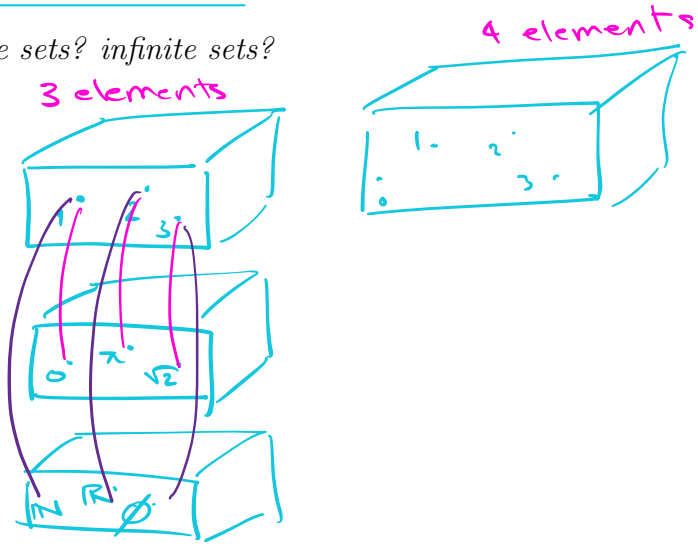
Definition: A **finite** set is one whose distinct elements can be counted by a natural number.

Motivating question: when can we say one set is *bigger than* another?

Does the definition change when we're comparing finite sets? infinite sets?

Which is bigger?

- The set $\{1, 2, 3\}$ or the set $\{0, 1, 2, 3\}$?
- The set $\{0, \pi, \sqrt{2}\}$ or the set $\{\mathbb{N}, \mathbb{R}, \emptyset\}$? Same size!
- The set $\mathbb{N}$ or the set $\mathbb{Q}$? Next week
- The set $\mathbb{Q}$ or the set $\mathbb{R}$? Next week
- The set $\mathbb{N}$ or the set $\mathbb{R}^+$? Next week



Key idea for cardinality: Counting distinct elements is a way of labelling elements with natural numbers. This is a function! In general, functions let us associate elements of one set with those of another. If the association is "*good*", we get a correspondence between the elements of the subsets which can relate the sizes of the sets.

Analogy: Musical chairs



People try to sit down when the music stops

Person☼ sits in Chair 1, Person☺ sits in Chair 2,

Person☹ is left standing!

What does this say about the number of chairs and the number of people?

Fewer chairs than people

More people than chairs

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Recall that a function is defined by its (1) domain, (2) codomain, and (3) rule assigning each element in the domain exactly one element in the codomain. The domain and codomain are nonempty sets. The rule can be depicted as a table, formula, English description, etc.

A function can *fail to be well-defined* if there is some domain element where the function rule doesn't give a unique codomain element. For example, the function rule might lead to more than one potential image, or to an image outside of the codomain.

Domain Codomain Rule

Example: $f_A : \mathbb{R}^+ \to \mathbb{Q}$ with $f_A(x) = x$ is **not** a well-defined function because

Counterexample: $\sqrt{2}$ is a positive real number (so in domain) but isn't rational (by Monday) so $f_A(\sqrt{2}) = \sqrt{2}$ is not in the codomain.

Domain Codomain rule

Example: $f_B : \mathbb{Q} \to \mathbb{Z}$ with $f_B\left(\frac{p}{q}\right) = p + q$ is **not** a well-defined function because

Counterexample: $0.5 = \frac{1}{2} = \frac{2}{4}$

with representation $\frac{1}{2}$ we get $f_B(0.5) = f_B(\frac{1}{2}) = 1+2 = 3$

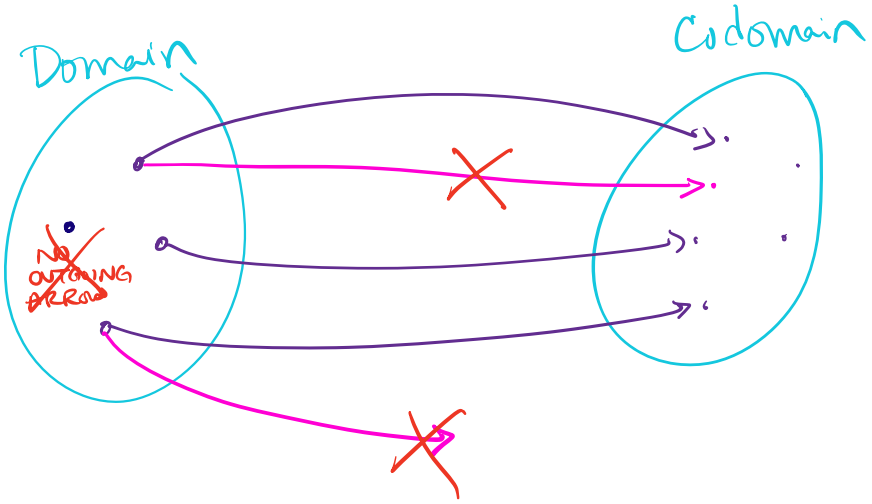
with representation $\frac{2}{4}$ we get $f_B(0.5) = f_B(\frac{2}{4}) = 2+4 = 6$

Domain Codomain rule

Example: $f_C : \mathbb{Z} \to \mathbb{R}$ with $f_C(x) = \frac{x}{|x|}$ is **not** a well-defined function because

Counterexample: 0

$f_C(0) = \frac{0}{|0|} = \frac{0}{0}$ undefined!



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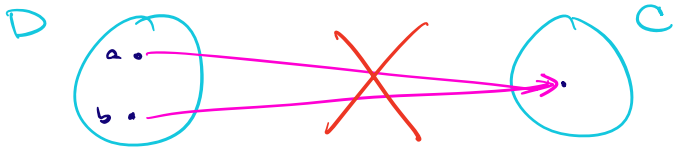
Additional properties functions may (or may not) have:

Definition : A function $f : D \to C$ is **one-to-one** (or injective) means for every a, b in the domain D, if $f(a) = f(b)$ then $a = b$.

Formally, $f : D \to C$ is one-to-one means $\forall a \in D \forall b \in D (f(a) = f(b) \to a = b)$

Informally, a function being one-to-one means "no duplicate images".

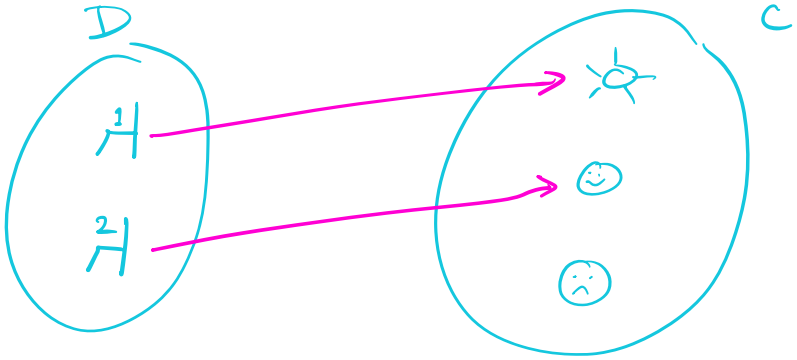
Contrapositive: $\forall a \in D \forall b \in D (a \neq b \to f(a) \neq f(b))$

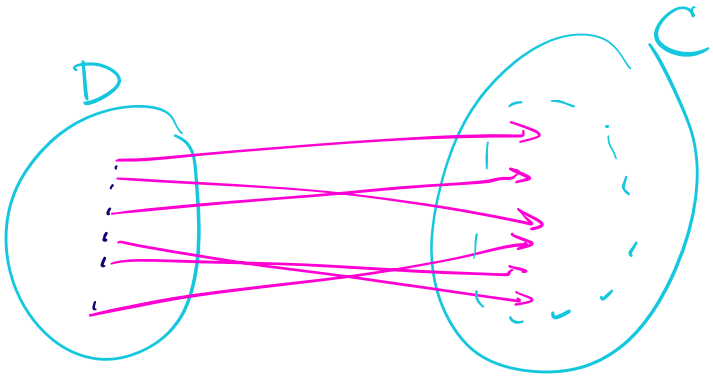


Definition: For nonempty sets A, B, we say that **the cardinality of A is no bigger than the cardinality of B**, and write $|A| \leq |B|$, to mean there is a one-to-one function with domain A and codomain B. Also, we define $|\emptyset| \leq |B|$ for all sets B.

In the analogy: The function *sitter* : $\{Chair1, Chair2\} \to \{Person☼, Person☺, Person☹\}$ given by $sitter(Chair1) = Person☼$, $sitter(Chair2) = Person☺$, is one-to-one and witnesses that

$$|\{Chair1, Chair2\}| \leq |\{Person☼, Person☺, Person☹\}|$$



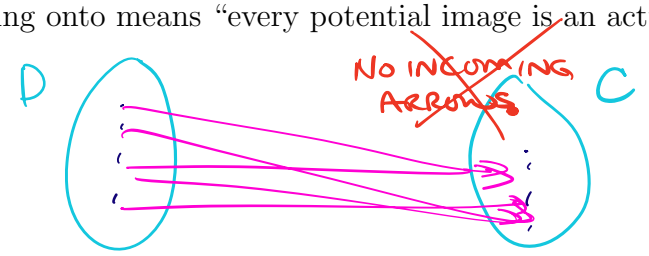


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Definition: A function $f : D \to C$ is **onto** (or surjective) means for every b in the codomain, there is an element a in the domain with $f(a) = b$.

Formally, $f : D \to C$ is onto means $\forall b \in C \exists a \in D\ (f(a) = b)$

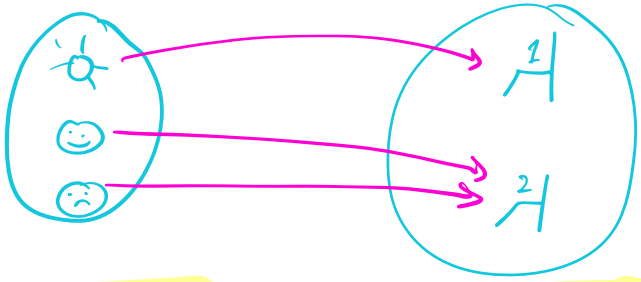
Informally, a function being onto means "every potential image is an actual image".



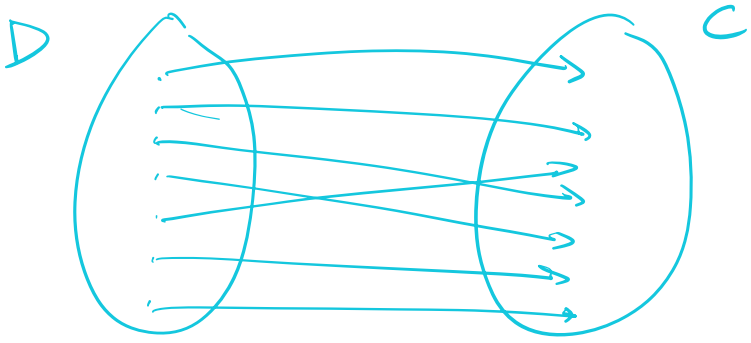
Definition: For nonempty sets A, B, we say that **the cardinality of A is no smaller than the cardinality of B**, and write $|A| \geq |B|$, to mean there is an onto function with domain A and codomain B. Also, we define $|A| \geq |\emptyset|$ for all sets A.

In the analogy: The function *triedToSit* : $\{Person☼, Person☺, Person☹\} \to \{Chair1, Chair2\}$ given by $triedToSit(Person☼) = Chair1$, $triedToSit(Person☺) = Chair2$, $triedToSit(Person☹) = Chair2$, is onto and witnesses that

$$|\{Person☼, Person☺, Person☹\}| \geq |\{Chair1, Chair2\}|$$



Definition : A function $f : D \to C$ is a **bijection** means that it is both one-to-one and onto. The **inverse** of a bijection $f : D \to C$ is the function $g : C \to D$ such that $g(b) = a$ iff $f(a) = b$.



Relabelling

For nonempty sets A, B we say

$|A| \leq |B|$ means there is a one-to-one function with domain A, codomain B
$|A| \geq |B|$ means there is an onto function with domain A, codomain B
$|A| = |B|$ means there is a bijection with domain A, codomain B

For all sets A, we say $|A| = |\emptyset|$, $|\emptyset| = |A|$ if and only if $A = \emptyset$.

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